

$$\cdot \mathcal{L} \{ e^{bt} \cdot u(t) \}$$

$$= \int_0^{\infty} e^{st} \cdot e^{-bt} dt = \int_0^{\infty} e^{t(b-s)} dt$$

$$= \int_0^{\infty} e^{-t(s-b)} dt \quad ; \quad \begin{aligned} u &= -t(s-b) \\ du &= -(s-b) \cdot dt \end{aligned}$$

$$= \int_0^{\infty} e^u \cdot \frac{-1}{s-b} du \quad dt = -\frac{1}{s-b} du$$

$$= -\frac{1}{s-b} \left[ e^u \right]_0^{\infty} = -\frac{1}{s-b} \left[ e^{-\infty} - e^0 \right]$$

$$= -\frac{1}{s-b} [0 - 1] = \frac{1}{s-b} \quad \text{✓}$$



Región de Convergencia:

$$X(s) = \frac{1}{s-b} ; e^{-t(s-b)} \rightarrow e^{-t(\sigma + j\omega - b)}$$
$$e^{-t(\sigma-b)} \cdot e^{-tj\omega}$$

Que pasa con  $b$ :

Senos y cosenos  
(complejos)

$$\bullet \sigma - b > 0 \rightarrow \sigma > b \quad \in [-1, 1]$$

$$e^{-\infty (+N)} \rightarrow 0$$

$$\bullet \sigma - b < 0 \rightarrow \sigma < b$$

$$e^{-\infty (-N)} = e^{\infty} \rightarrow \infty$$

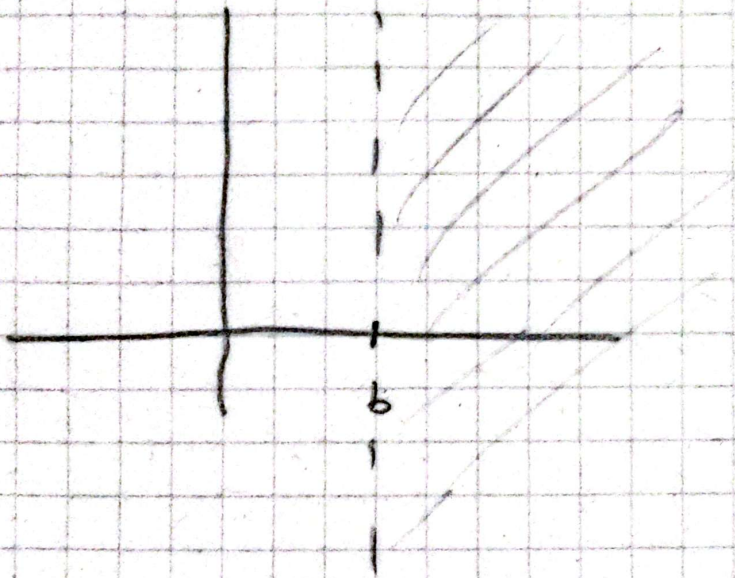
Nuestro ROC debe estar en:

$$\text{Parte real} \rightarrow \sigma - b > 0$$

Noes ha Parte real menos la constante  
real  $b$  de la  $s$  sea mayor a 0

$$\operatorname{Re}\{s\} - b > 0$$

$$\operatorname{Re}\{s\} > b$$





$$\bullet \mathcal{L} \{ \cos(\omega_0 t) \cdot u(t) \}$$

$$= \int_0^{\infty} \cos(\omega_0 t) \cdot e^{-st} dt$$

$$= \int_0^{\infty} \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} \cdot e^{-st} dt$$

$$= \frac{1}{2} \int_0^{\infty} e^{j\omega_0 t - st} + e^{-j\omega_0 t - st} dt$$

$$= \frac{1}{2} \int_0^{\infty} e^{-t(s-j\omega_0)} + e^{-t(s+j\omega_0)} dt$$

$$= \frac{1}{2} \left[ \int_0^{\infty} e^{-t(s-j\omega_0)} dt + \int_0^{\infty} e^{-t(s+j\omega_0)} dt \right]$$

$$= \frac{1}{2} \left[ \int_0^{\infty} e^{-t(s-j\omega_0)} dt + \int_0^{\infty} e^{-t(s+j\omega_0)} dt \right]$$

$$u = -t(s-j\omega_0)$$

$$du = -(s-j\omega_0) dt$$

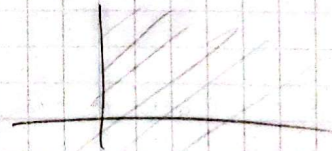
$$dt = -\frac{1}{s-j\omega_0} du$$

$$= \frac{1}{2} \left[ \frac{1}{j\omega_0 - s} \left[ e^{-t(s-j\omega_0)} \right]_0^{\infty} - \frac{1}{s+j\omega_0} \left[ e^{-t(s+j\omega_0)} \right]_0^{\infty} \right]$$

$$\text{Terms: } e^{-t(s+j\omega-j\omega_0)}$$

$$e^{-t\sigma} \cdot e^{-tj\omega} \cdot e^{tj\omega_0}$$

$$\bullet e^{-t\sigma} \rightarrow \operatorname{Re}\{s\} > 0$$



$$u = -t(s+j\omega_0)$$

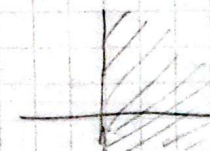
$$du = -(s+j\omega_0) dt$$

$$dt = -\frac{1}{s+j\omega_0} du$$

$$e^{-t(s+j\omega+j\omega_0)}$$

$$e^{-t\sigma} \cdot e^{-tj\omega} \cdot e^{-tj\omega_0}$$

$$\bullet e^{-t\sigma} \rightarrow \operatorname{Re}\{s\} > 0$$





Enonces:

$$= \frac{1}{2} \left[ \frac{1}{j\omega_0 - s} \left[ \cancel{e^{-s}} - \cancel{e^{-s}} \right] - \frac{1}{s + j\omega_0} \left[ \cancel{e^{-s}} - \cancel{e^{-s}} \right] \right]$$

$$= \frac{1}{2} \left[ \frac{1}{s - j\omega_0} + \frac{1}{s + j\omega_0} \right]$$

$$= \frac{1}{2} \left[ \frac{s + j\omega_0 + s - j\omega_0}{s^2 - (j\omega_0)^2} \right]$$

$$= \frac{1}{2} \left[ \frac{\cancel{2} s}{s^2 - (j\omega_0)^2} \right]$$

$$= \frac{s}{s^2 - (j\omega_0)^2}$$



$$\bullet \mathcal{L} \{ \sin(\omega_0 t) \cdot u(t) \}$$

$$= \int_0^{\infty} \sin(\omega_0 t) \cdot e^{-st} dt$$

$$= \int_0^{\infty} \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j} \cdot e^{-st} dt$$

$$= \frac{1}{2j} \int_0^{\infty} e^{-t(s-j\omega_0)} - e^{-t(s+j\omega_0)} dt$$

$$u = -t(s-j\omega_0)$$

$$du = -(s-j\omega_0) dt$$

$$dt = \frac{-1}{s-j\omega_0} du$$

$$u = -t(s+j\omega_0)$$

$$du = -(s+j\omega_0) dt$$

$$dt = -\frac{1}{s+j\omega_0} du$$

Con misma ROC definida

$$= \frac{1}{2j} \left[ -\frac{1}{s-j\omega_0} \left( e^{-t(s-j\omega_0)} \right)_0^{\infty} + \frac{1}{s+j\omega_0} \left( e^{-t(s+j\omega_0)} \right)_0^{\infty} \right]$$

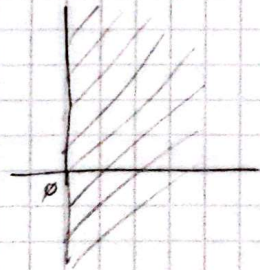
Termos:

$$-t(\sigma + j\omega - j\omega_0)$$

$$e^{-t\sigma} \cdot e^{-tj\omega} \cdot e^{tj\omega_0}$$

$$\in [-1, 1]$$

$$\text{Re}\{s\} > 0$$

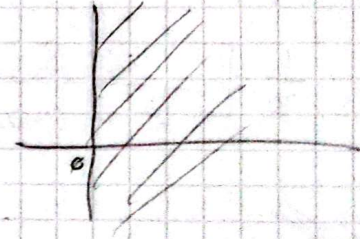


$$-t(\sigma + j\omega + j\omega_0)$$

$$e^{-t\sigma} \cdot e^{-tj\omega} \cdot e^{-tj\omega_0}$$

$$\in [-1, 1]$$

$$\text{Re}\{s\} > 0$$



$$= \frac{1}{2j} \left[ -\frac{1}{s-j\omega_0} \left( e^{-\infty} - e^0 \right) + \frac{1}{s+j\omega_0} \left( e^{-\infty} - e^0 \right) \right]$$



$$= \frac{1}{2j} \left[ \frac{1}{s - j\omega_0} - \frac{1}{s + j\omega_0} \right]$$

$$= \frac{1}{2j} \left[ \frac{\cancel{s + j\omega_0} - \cancel{s + j\omega_0}}{s^2 - (j\omega_0)^2} \right]$$

$$= \frac{1}{\cancel{2j}} \left[ \frac{\cancel{2j}\omega_0}{s^2 - (j\omega_0)^2} \right]$$

$$= \frac{\omega_0}{s^2 - (j\omega_0)^2} = \frac{\omega_0}{s^2 - (-1)\omega_0^2}$$

$$\checkmark = \frac{\omega_0}{s^2 + \omega_0^2}$$



$$\mathcal{L} \{ e^{bt} \cdot \cos(\omega_0 t) \cdot u(t) \}$$

$$= \int_0^{\infty} e^{bt} \cdot \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} e^{-st} dt$$

$$= \int_0^{\infty} \frac{1}{2} \cdot e^{-t(s-b-j\omega_0)} + e^{-t(s-b+j\omega_0)} dt$$

$$= \frac{1}{2} \int_0^{\infty} e^{-t(s-(b+j\omega_0))} + e^{-t(s-(b-j\omega_0))} dt$$

$$u = -t(s-(b+j\omega_0))$$

$$u = -t(s-(b-j\omega_0))$$

$$du = -(s-(b+j\omega_0)) dt$$

$$du = -(s-(b-j\omega_0)) dt$$

$$dt = -\frac{1}{s-(b+j\omega_0)} du$$

$$dt = -\frac{1}{s-(b-j\omega_0)} du$$

$$= \frac{1}{2} \left[ -\frac{1}{s-(b+j\omega_0)} \left( e^{-t(s-(b+j\omega_0))} \right)_0^{\infty} - \dots \right]$$

$$\dots - \frac{1}{s-(b-j\omega_0)} \left( e^{-t(s-(b-j\omega_0))} \right)_0^{\infty} \right]$$

La Rac Seri :

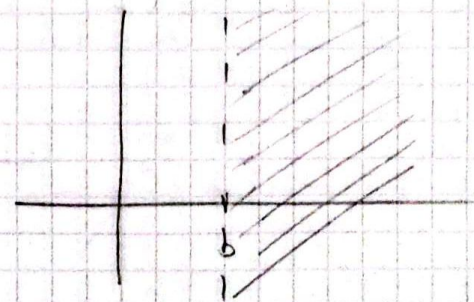
$$\bullet e^{-t(s-(b+j\omega_0))}$$

$$e^{-t(s+j\omega-b-j\omega_0)}$$

$$e^{-t(s-b)} \cdot e^{-tj(\omega-\omega_0)}$$

$$\in [-1, 1]$$

$$\circ s-b > 0 \rightarrow \operatorname{Re}\{s\} > b$$





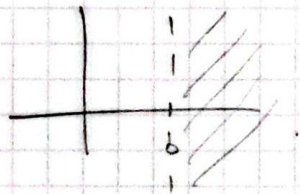
$$e^{-t(s - (b - j\omega_0))}$$

$$e^{-t(s + j\omega - b + j\omega_0)}$$

$$e^{-t(\sigma - b)} \cdot e^{-tj(\omega + \omega_0)}$$

$\in [-1, 1]$

$$\sigma - b > 0 \rightarrow \operatorname{Re}\{s\} > b$$



Teniendo clara nuestra región de convergencia:

$$= \frac{1}{2} \left[ -\frac{1}{s - (b + j\omega_0)} (e^{-\infty} - e^0) - \frac{1}{s - (b - j\omega_0)} (e^{-\infty} - e^0) \right]$$

$$= \frac{1}{2} \left[ \frac{1}{s - (b + j\omega_0)} + \frac{1}{s - (b - j\omega_0)} \right]$$

$$= \frac{1}{2} \left[ \frac{s - (b - j\omega_0) + s - (b + j\omega_0)}{s^2 - s(b - j\omega_0) - s(b + j\omega_0) + (b^2 + \omega_0^2)} \right]$$

$$= \frac{1}{2} \left[ \frac{\cancel{s} - b + \cancel{j\omega_0} + \cancel{s} - b - \cancel{j\omega_0}}{s^2 - sb + \cancel{j\omega_0} - sb - \cancel{j\omega_0} + (b^2 + \omega_0^2)} \right]$$

$$= \frac{1}{2} \left[ \frac{2s - 2b}{s^2 + (b^2 + \omega_0^2) - 2sb} \right]$$

$$= \frac{1}{2} \left[ \frac{\cancel{2}(s - b)}{(s^2 - 2sb + b^2) + \omega_0^2} \right]$$

$$(s - b)^2$$

$$= \frac{s - b}{(s - b)^2 + \omega_0^2}$$



$$\bullet \mathcal{L} \{ e^{bt} \sin(\omega_0 t) u(t) \}$$

$$= \frac{1}{2j} \left[ -\frac{1}{s-(b+j\omega_0)} \left( e^{-t(s-(b+j\omega_0))} \right)_0^\infty + \dots \right]$$

$$= \int_0^\infty e^{bt} \cdot \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j} \cdot e^{-st} dt$$

$$\dots + \frac{1}{s-(b-j\omega_0)} \left( e^{-t(s-(b-j\omega_0))} \right)_0^\infty \right]$$

$$= \int_0^\infty \frac{1}{2j} \left[ e^{-t(s-b-j\omega_0)} - e^{-t(s-b+j\omega_0)} \right] dt$$

Nu ma ROC scrie:

$$\bullet e^{-t(s-(b+j\omega_0))}$$

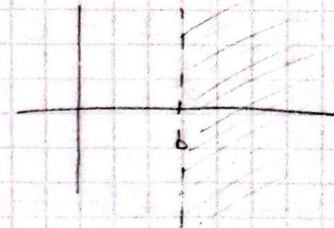
$$e^{-t(\sigma+j\omega-b-j\omega_0)}$$

$$e^{-t(\sigma-b)} \cdot e^{-tj(\omega-\omega_0)}$$

$$\in [-1, 1]$$

$$\sigma-b > 0$$

$$\Re\{s\} > b$$



$$= \frac{1}{2j} \int_0^\infty e^{-t(s-(b+j\omega_0))} - e^{-t(s-(b-j\omega_0))} dt$$

$$u = -t(s-(b+j\omega_0))$$

$$u = -t(s-(b-j\omega_0))$$

$$du = -(s-(b+j\omega_0)) \cdot dt$$

$$du = -(s-(b-j\omega_0)) \cdot dt$$

$$dt = -\frac{1}{s-(b+j\omega_0)} du$$

$$dt = -\frac{1}{s-(b-j\omega_0)} du$$



$$\bullet e^{-t(s - (b - j\omega_0))}$$

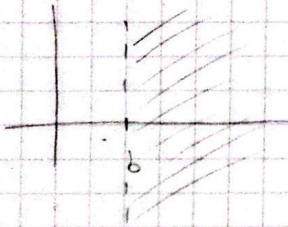
$$e^{-t(s + j\omega - b + j\omega_0)}$$

$$e^{-t(\sigma - b)} \cdot e^{-tj(\omega + \omega_0)}$$



$$e \in [-1, 1]$$

$$\sigma - b > 0 \rightarrow \text{Re}\{s\} > b$$



Temendo clara nossa ROC:

$$= \frac{1}{2j} \left[ -\frac{1}{s - (b + j\omega_0)} \left( e^{-\infty} - e^{0} \right) + \frac{1}{s - (b - j\omega_0)} \left( e^{-\infty} - e^{0} \right) \right]$$

$$= \frac{1}{2j} \left[ \frac{1}{s - (b + j\omega_0)} - \frac{1}{s - (b - j\omega_0)} \right]$$

$$= \frac{1}{2j} \left[ \frac{\cancel{s - b} + j\omega_0 - \cancel{s + b} + j\omega_0}{s^2 - s(b - j\omega_0) - s(b + j\omega_0) + (b^2 - j^2\omega_0^2)} \right]$$

$$= \frac{1}{2j} \left[ \frac{2j\omega_0}{s^2 - \cancel{s}b + \cancel{s}\omega_0 - \cancel{s}b - \cancel{s}\omega_0 + b^2 - \cancel{j^2}\omega_0^2} \right]$$

$$= \frac{1}{2j} \left[ \frac{2j\omega_0}{s^2 - 2sb + b^2 + \omega_0^2} \right]$$

$$= \cancel{\frac{1}{2j}} \frac{\omega_0}{(s - b)^2 + \omega_0^2}$$